1. What threshold did you set for blur? How did you decide (trial and error? what did you test on)?

A) Threshold Used

The blur threshold was set to **300**, and the variance of the Laplacian operator was used as the blur measurement.

"is_blurry": blur_score < 300

How the Threshold Was Determined

The threshold was determined through an iterative trial-and-error process. Images from four categories were collected:

- Good-quality fingerprint images
- Blurred fingerprint images
- Dark fingerprint images
- Images affected by glare

The blur score was calculated for every image, and the results were compared. Clear fingerprint images consistently produced higher scores, whereas blurred images produced much lower values. After several experiments, a threshold value of 300 provided the best separation between acceptable and unacceptable images.

This approach helped minimize false acceptance and false rejection rates.

2. Which metric was hardest to implement correctly? What went wrong first?

A) Glare Detection

Glare detection was the most challenging part of the project. Initially, the algorithm identified glare by counting the number of pixels above a fixed intensity threshold. However, the results were inconsistent because reflections from the phone flash were spread across the fingertip instead of appearing in a single location.

During testing, several problems were observed:

- Dark images were sometimes classified incorrectly.
- Some glare images were identified as blurred images.
- Excessive illumination reduced ridge visibility.
- Different skin tones produced different intensity patterns.

To improve the performance, Gaussian filtering and intensity-based analysis were incorporated before calculating the glare ratio. Although the method is not perfect, the final quality gate successfully rejects poor-quality images.

3. What is NFIQ2? Why is a score designed for contact scanners not reliable for phone camera images?

A) NFIQ2 (NIST Fingerprint Image Quality 2) is a fingerprint quality assessment framework developed by NIST to evaluate whether a fingerprint image is suitable for biometric matching. It analyzes factors such as ridge clarity, contrast, orientation consistency, and local image quality.

Why It Is Less Suitable for Smartphone Images

NFIQ2 was designed primarily for contact-based fingerprint scanners operating in controlled environments. Smartphone images, however, introduce several additional challenges:

- Uneven illumination
- Motion blur
- Perspective distortion
- Variable image resolution
- Different capture distances
- Background interference

Because of these factors, NFIQ2 scores may not accurately represent the quality of contactless fingerprint images captured using mobile devices.

4. Name 3 other quality problems you'd add checks for in a real deployment (e.g., wet finger, wrong angle, finger too far from camera).

A)

- Wet Finger Detection: Identifies moisture that can create reflections and hide ridge patterns.
- Finger Orientation Detection: Ensures that the finger is positioned correctly to improve matching accuracy.
- Distance Estimation: Checks whether the finger is too close to or too far from the camera.

Other possible improvements include dry skin detection, motion analysis, and background segmentation.

5. If a rural agricultural worker's fingerprints are naturally worn and give consistently poor ridge clarity scores, what should the system do differently for them?

A) People working in agriculture, construction, and similar occupations often have naturally worn fingerprints, which can lead to low ridge clarity scores. Rejecting these users solely on the basis of fingerprint quality would reduce the accessibility and reliability of the system.

Instead, the system should adopt a more flexible approach by:

- Using adaptive threshold values.
- Capturing multiple fingerprint samples.
- Combining information from different fingers.
- Applying enhancement techniques such as CLAHE and Gabor filtering.
- Incorporating other biometric methods, such as iris recognition or facial recognition, when necessary.

This approach improves accuracy, fairness, and the overall robustness of the biometric system.